

ICU MASTERCLASS— INOTROPES AND VASOPRESSORS



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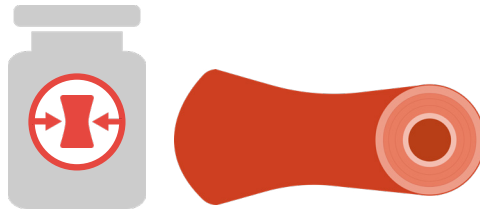
Chapter 1

PHYSIOLOGIC EFFECTS OF VASOACTIVE AGENTS



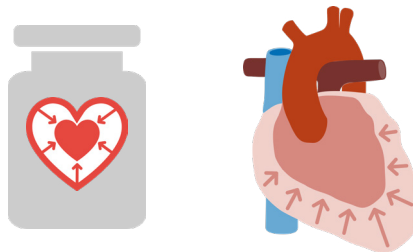
Introducing inotropes and vasopressors

Vasopressor—or *pressor*, refers to an agent that causes vascular constriction, either venous or arterial. These may also be called vasoconstrictors.



Vaso = vessel

Inotrope—an agent that affects the force of the heart's contraction.



In = fiber / sinew; tropos = turn / move

A positive inotrope is an agent that promotes inotropy, meaning that it increases contractility of the heart. Epinephrine and other beta-agonists are examples of positive inotropes.

Agents with the opposite effect, leading to decreased contractility, are referred to as negative inotropes. Beta-blockers like metoprolol are negative inotropes.

Sometimes terms will be joined and an agent with combined actions may be called an inodilator or an inopressor.



Inotropy should not be confused with ionotropy, which means related to ion channels.

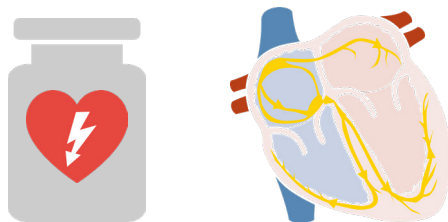
Chronotropic agents—increase heart rate by affecting the electrical system and the neuronal stimulation to the heart.



Chrono = time; tropos = turn / move

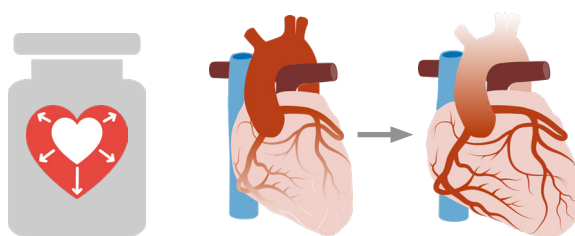
Most agents that are positive inotropes (e.g., epinephrine) are also positive chronotropes. Agents which slow the heart rate (e.g., beta-blockers) are known as negative chronotropes and are usually also negative inotropes.

Dromotropic agents—increase the conduction speed in the heart and the AV node. An example is epinephrine.



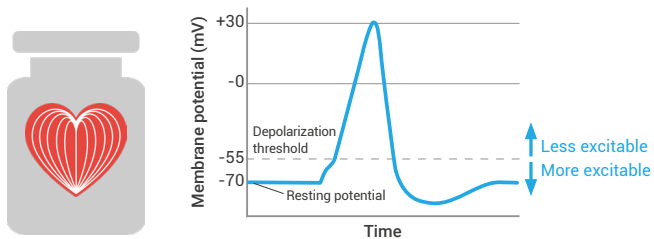
Dromo = running a race / course; tropos = turn / move

Lusitropic agents—increase diastolic relaxation. Catecholamines promote diastolic relaxation, which is largely a calcium-mediated process. Decreased calcium efflux from cells during repolarization will decrease lusitropy.



Lusis = loosen / relax; tropos = turn / move

Bathmotropy—refers to myocardial excitability. Positive bathmotropic agents (e.g., epinephrine) make the myocardium easier to depolarize, by bringing the resting membrane potential and the depolarization threshold closer together. Negative bathmotropic conditions (e.g., hypokalemia) move the resting membrane potential farther from the depolarization threshold.



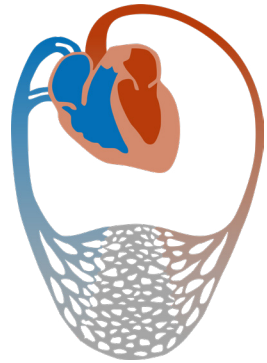
Bathmos = step / threshold

Differentiating the types of shock

In the normal situation, oxygen-rich blood flows down a pressure gradient from the aorta through the capillary beds, toward the venous circulation, where it delivers oxygen.

Shock is defined by inadequate organ perfusion.

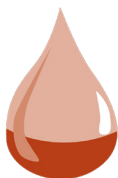
Assessing perfusion can be challenging because we really don't have great methods to assess most organ functions rapidly. Instead, we rely on surrogate measurements like blood pressure or cardiac output, which represents blood flow from the heart.



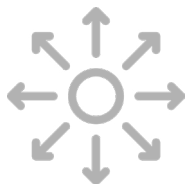
However, it is possible to have adequate blood flow with a low pressure (e.g., in a resting state). It is also possible to have a normal cardiac output but inadequate organ perfusion (e.g., in a hypermetabolic state or with vasoconstriction in the distal vessels).

If we focus on where the problem lies in this pathway, we can have a systematic and nearly universal approach to helping a patient with inadequate organ perfusion. Is there not enough preload? Is there inadequate afterload (low systemic vascular resistance)? Is there a problem with the heart itself?

Four main classes of shock



Hypovolemic



Distributive



Cardiogenic



Obstructive

Treatment may vary depending on etiology

- Volume resuscitation for hypovolemia
- Increased vasoconstriction for distributive shock
- Inotropy or interventions to improving the heart's pumping function for cardiogenic shock
- Relieving an obstructive cause of shock (e.g., cardiac tamponade)

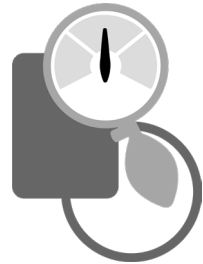
Often, more than one of these derangements is present concomitantly.

Performing vasoactive monitoring

Although it is difficult to measure perfusion, there are several other pieces of information we can use to guide our therapies.

Blood pressure

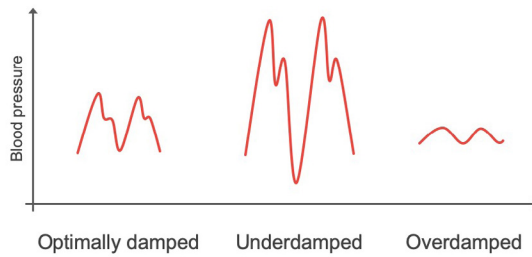
This is one of the most basic elements of blood flow that is readily measurable. If there is inadequate pressure to force blood through the capillaries, there will be inadequate perfusion.



However, pressure is not the only factor at play here. Constricted capillaries with a lot of resistance will need more pressure to create the same amount of flow, and by the same token, dilated capillaries with low resistance will not need as much pressure to sustain blood flow.

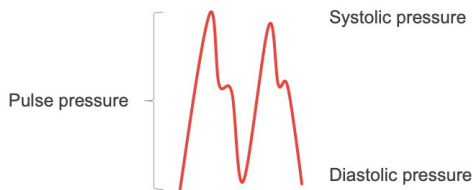
We can measure blood pressure intermittently with noninvasive sphygmomanometers or continuously with an intravascular pressure monitor (usually called an arterial line). Obviously, continuous information improves our ability to titrate interventions but this must be balanced with the risks of the procedure including pain, infection or injury to the vasculature or surrounding structures.

Arterial pressure monitoring may be confounded by over- or under-dampening of the pressure conduit to the sensor. This may be recognized by an increased or decreased pulse pressure respectively and the square wave test can help distinguish this.



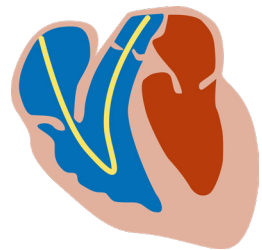
Pulse pressure

The difference between systolic and diastolic blood pressure (the pulse pressure) can give an idea of the relative strength of pulses. Greater pulse pressure results from stronger contractions from the heart. It is subject to many confounding factors but a trend may still be useful.



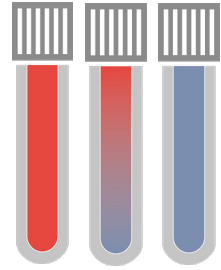
Invasive pulmonary artery catheters

Perhaps even more important than pressure, we can also try to look at blood flow (cardiac output) through a variety of methods. Invasive pulmonary artery catheters may provide estimates of cardiac output by measuring temperature changes using an intravascular thermistor. However, this is subject to error, especially in patients with tricuspid regurgitation or intracardiac shunting. The procedure may also be technically challenging and may expose the patient to risks of complications.



Oxygen saturation

Measurement of the oxygen saturation of central or mixed venous blood samples may provide an estimate of cardiac output using Fick's law. However, this is also subject to error, mainly based on assumptions about oxygen consumption and it requires access to central or pulmonary arterial blood.



Echocardiography

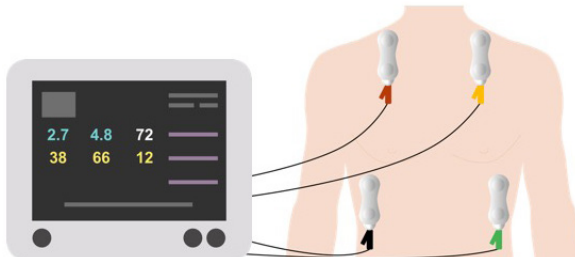
Echocardiography may provide estimates of blood flow noninvasively but this is subject to operator error and availability of adequate imaging. It is also difficult to monitor continuously.



Non-invasive cardiac output monitoring

Other forms of non-invasive cardiac output monitoring (NICOM) have emerged as well. These are based on various different measurements including pulse contour analysis, bioimpedance, and phase-shifting, each subject to their own limitations.

Each method of measurement has its own advantages and limitations. Being familiar with several techniques will allow flexibility, as some methods may be better suited for specific scenarios.

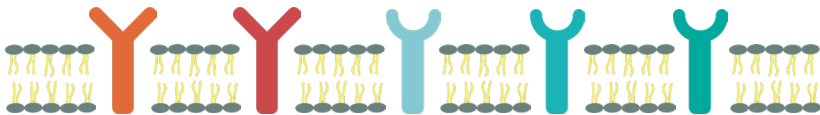


Deciphering vasoactive receptors

There are a variety of vasoactive receptors throughout the body. An understanding of the physiologic basis for the effects of our interventions is essential to tailor appropriate therapy.

Catecholamine receptors

Catecholamine receptors are broadly classified as alpha and beta receptors, each with their own subtypes, and with many different functions. We will focus on the most hemodynamically significant.



Alpha 1 agonism

- Promotes smooth muscle contraction
- Promotes vasoconstriction
- Increases arterial pressure
- Causes sphincter contraction in gastrointestinal tract and bladder

Alpha 2 agonism

- May lower blood pressure
- May lower heart rate
- May alter neurotransmitter function (especially norepinephrine)

Beta 1 agonism

- Increases heart rate (positive inotrope)
- Increases contractility (positive chronotrope)
- Increases cardiac output (dromotrope)

Beta 2 agonism

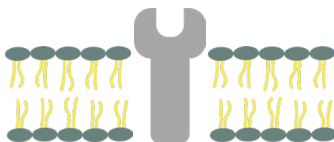
- Promotes smooth muscle relaxation
- Lowers intravascular pressure

Beta 3 agonism

- Increases lipolysis
- Promotes bladder relaxation
- No major hemodynamic effects

Dopamine receptors

Several dopamine receptor subtypes exist throughout the body with various functions. Primarily in the brain, dopamine functions as a neurotransmitter and modulates neuroendocrine functions. However, dopamine receptors are also present in the vasculature.

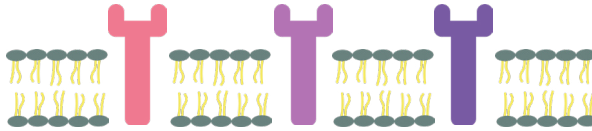


Dopamine receptor agonism

- Promotes vasodilation
- Promotes diuresis (because of vasodilation of renal arteries)

Vasopressin receptors

There are three types of vasopressin receptors identified.



V1 receptors (which were previously called V1A) exist on vascular smooth muscle and platelets.

V1 agonism

- Promotes vasoconstriction
- No marked effect on the pulmonary vasculature
 - increase systemic blood pressure without raising pulmonary vascular pressures
- May promote thrombosis

V2 receptors are present in the renal collecting ducts and endothelial cells.

V2 agonism

- Promotes an antidiuretic effect



Another name for vasopressin is antidiuretic hormone (ADH).

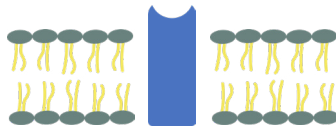
V3 receptors (previously called V1B receptors) exist on the anterior pituitary gland.

V3 agonism

- Promotes release of corticotropin
 - affects hemodynamics through a steroid-mediated response

Angiotensin II receptors

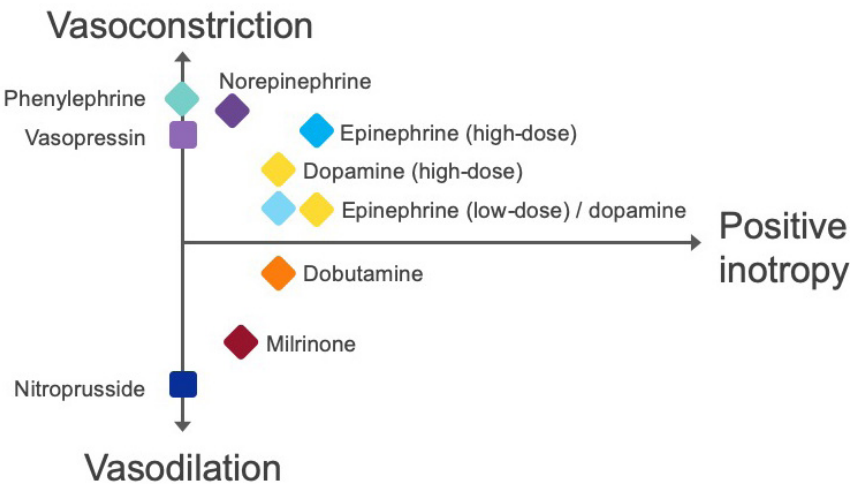
Although several types of angiotensin II receptors have been identified, only the function of the type 1 receptor is well understood. It exists in the heart, blood vessels, kidney, adrenal cortex, lungs, and brain.



Angiotensin II receptor (type I) agonism

- Promotes vasoconstriction
- Promotes aldosterone production
- Promotes vasopressin release
- Decreases renal blood flow

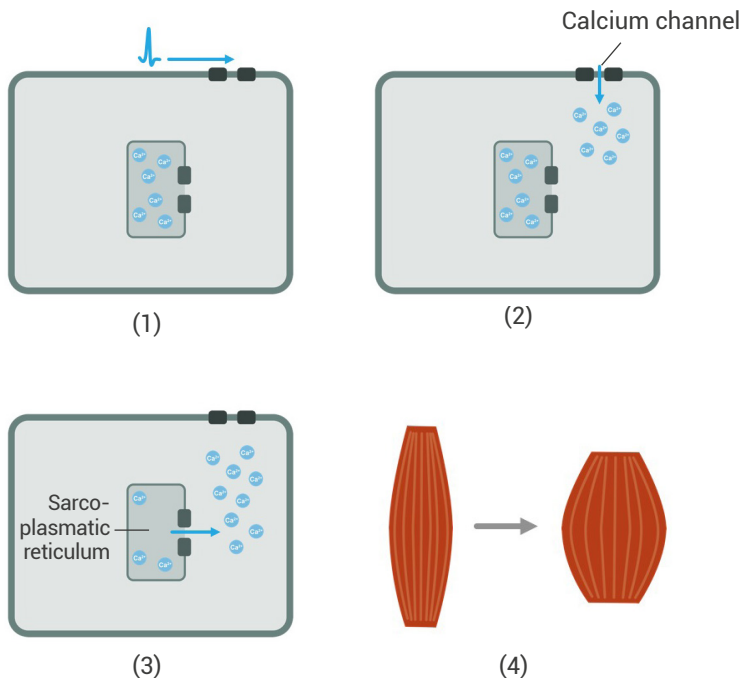
Summary



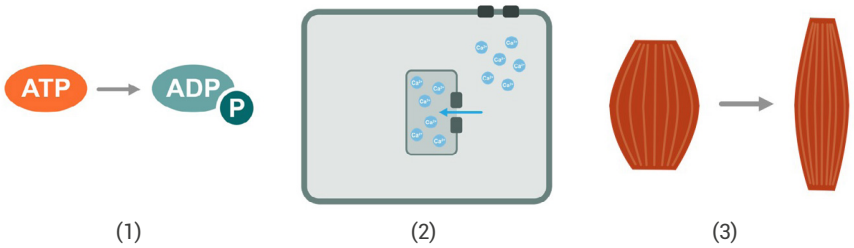
Appreciating myocyte contraction

There are several pathways involved in myocyte contraction, which ultimately drives cardiac output and thus perfusion.

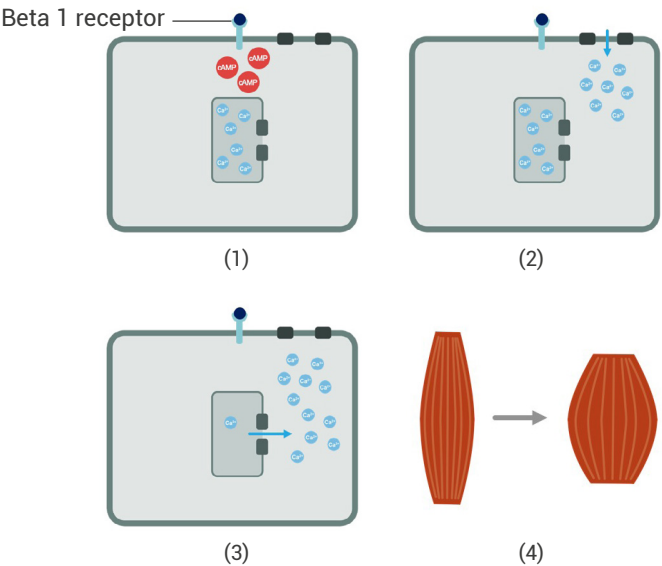
Depolarization of the myocyte membrane (1) causes calcium influx through the L-type calcium channel (2). This increase in intracellular calcium causes release of additional calcium from the ryanodine receptors on the sarcoplasmic reticulum (3). This increase in calcium concentration stimulates myocyte contraction (4).



Cleavage of ATP (1) and a decrease in calcium, as it returns to the sarcoplasmic reticulum and leaves the cell (2), allows relaxation of the myocyte (3).

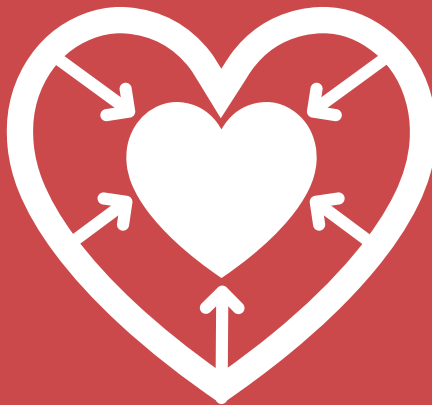


Activation of the beta 1 receptor on the surface of the myocyte increases the presence of intracellular cyclic AMP (1). This, in turn, potentiates the action of the calcium channel (2), increasing the influx of calcium into the cell (3), and increasing contractility (4).



Chapter 2

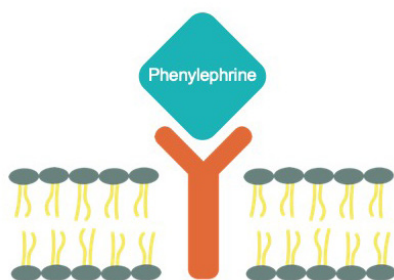
INOTROPES AND VASOPRESSORS



Mastering catecholamine-based vasopressors

Phenylephrine

Phenylephrine is a vasopressor which works by stimulating alpha 1-adrenergic receptors in the peripheral vasculature.

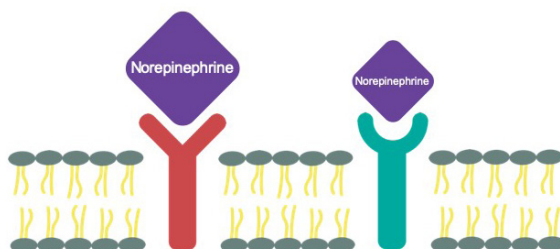


It does not stimulate any other receptors, thus does not induce tachycardia (in contrast to norepinephrine, which has some beta agonism). In fact, it will often result in a lower heart rate reflexively as the blood pressure increases. However, for this reason, it may also decrease cardiac output by increasing afterload and does cause pulmonary vasoconstriction, which may be particularly deleterious for the right ventricle.

It is generally dosed from about 10–200 μg / min.

Norepinephrine

Norepinephrine, or noradrenaline, is a potent alpha-adrenergic agonist, which also has milder beta-adrenergic agonism.



It has demonstrated advantages over other vasopressors for a variety of conditions including undifferentiated shock and septic shock.

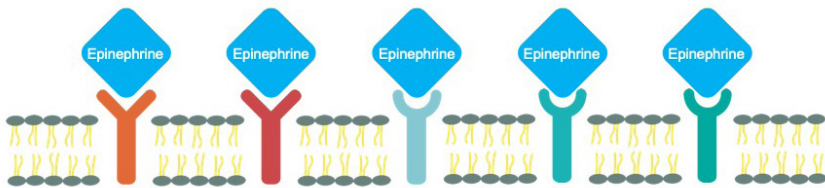
Norepinephrine is a very potent vasoconstrictor; therefore, it has the risk of tissue necrosis if it extravasates, so central venous administration is recommended when available.

Its dosing range is commonly $0.01-3 \mu\text{g} / \text{kg} / \text{min}$ or $1-30 \mu\text{g} / \text{min}$.

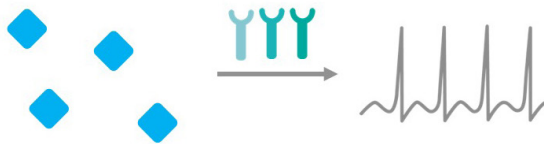
Grasping catecholamine-based inotropes

Epinephrine

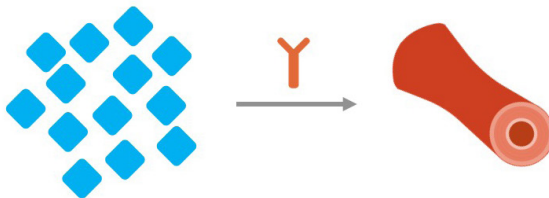
Epinephrine, or adrenaline, is a naturally occurring agonist for all types of adrenergic receptors.



At lower concentrations, the beta adrenergic effects predominate, which results in an increase in heart rate and contractility from the beta 1 receptor, and smooth muscle relaxation from the beta 2 receptor.



With increasing doses, more alpha 1 agonism is present, leading to vasoconstriction.



Epinephrine has the advantages of being multimodal in this fashion, and is well tolerated in the subcutaneous tissue if it extravasates.

However, its dose-dependent response may be variable and unpredictable, and it does cause increased lactate production and release, which may confound some methods to measure perfusion.

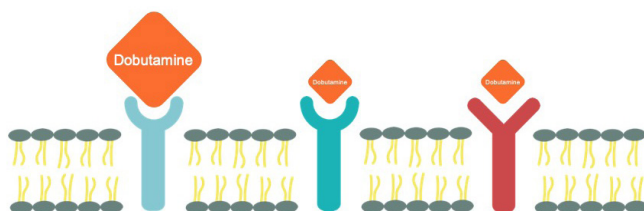
The typical dosing range for an epinephrine infusion is 0.01–0.3 $\mu\text{g} / \text{kg} / \text{min}$.

Dobutamine

Dobutamine is a synthetic beta-agonist.

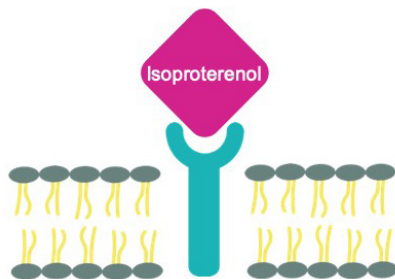
It preferentially activates beta 1 receptors with about three times the affinity that it has for beta 2 receptors. It also has mild alpha-agonist activity, thus it may cause less hypotension than other inodilators.

Dobutamine is typically dosed from 2–15 $\mu\text{g} / \text{kg} / \text{min}$, though higher doses are occasionally used.



Isoproterenol

Isoproterenol is a synthetic nonselective beta-agonist.



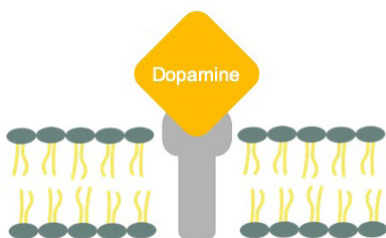
It increases inotropy, and has a more profound chronotropic effect than dobutamine. It also causes more vasodilation and hypotension as a result of its less selective beta 2 agonism, and has almost no alpha activity.

Isoproterenol is generally dosed at 0.5–5 μg / min, although doses up to 20 μg / min are common for severe bradycardias.

Appreciating indirect-acting agents

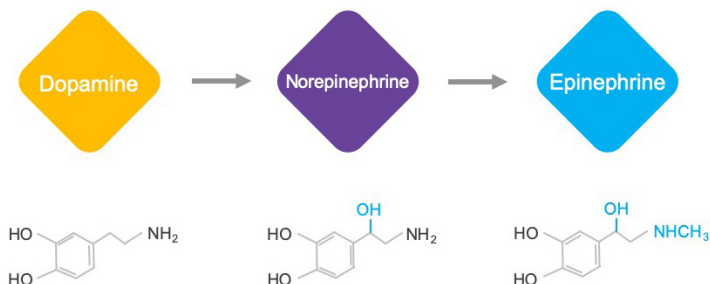
Dopamine

Dopamine functions primarily as a neurotransmitter. However, its role as a vasoactive agent outside of the brain has been recognized and used therapeutically since the 1960's.



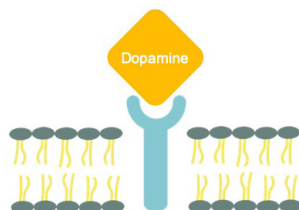
It acts directly on dopamine receptors in the renal, mesenteric, and coronary vascular beds causing vasodilation.

However, the majority of its hemodynamic effects are a result of conversion into norepinephrine and epinephrine. Increasing doses of dopamine lead to increased production of norepinephrine, and subsequently epinephrine, which lead to vasoconstriction and inotropy.

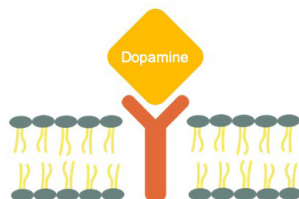


At lower doses, usually 1–5 $\mu\text{g} / \text{kg} / \text{min}$, dopamine leads to stimulation of primarily dopamine. This leads to an increase in renal blood flow and inhibition of aldosterone, which increases diuresis and natriuresis.

From 5–15 $\mu\text{g} / \text{kg} / \text{min}$, dopamine and its metabolites begin to have more effect on beta receptors, increasing heart rate and contractility.



At doses over 15 $\mu\text{g} / \text{kg} / \text{min}$, it leads to increased activation of alpha 1 receptors, leading to vasoconstriction and increasing blood pressure.



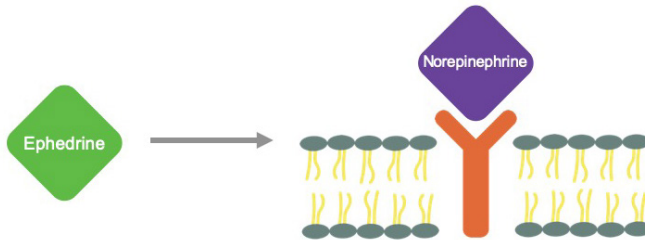
These ranges are approximate and these effects occur at a variable rate, thus dosing can be unpredictable. Even within the same patient, this may change over time. A stable physiologic effect is difficult to achieve.

Dopamine increases diuresis and natriuresis (even at low doses) by promoting renal blood flow and inhibiting aldosterone release. However, this has repeatedly failed to translate into a clinically important protective effect on the kidneys.

For many of these reasons, dopamine is no longer considered a first-line agent for most shock scenarios, if alternative agents are available.

Ephedrine

Ephedrine is a sympathomimetic amine, which was first used therapeutically in 1926.



Its hemodynamic effects occur primarily as a result of stimulation of norepinephrine release, so its clinical effects are similar to low doses of epinephrine or norepinephrine. It may be less effective in patients who are experiencing catecholamine depletion (e.g., in a prolonged period of critical illness).

Ephedrine is generally dosed as a bolus, either 5–10 mg IV or 25–50 mg intramuscularly (IM).

Reviewing vasopressin

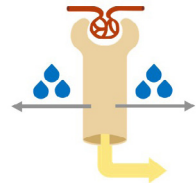
Vasopressin

Vasopressin is a naturally occurring nona-peptide released from the posterior pituitary gland in response to elevated osmotic pressure, which gives rise to its other name antidiuretic hormone (ADH).



In humans, it contains arginine in the eighth position and thus may also be abbreviated as AVP (arginine vasopressin) to distinguish it from other forms of vasopressin in other animals.

The body's response to vasopressin does several things. First, it causes increased free water resorption from the collecting tubules in the kidney.



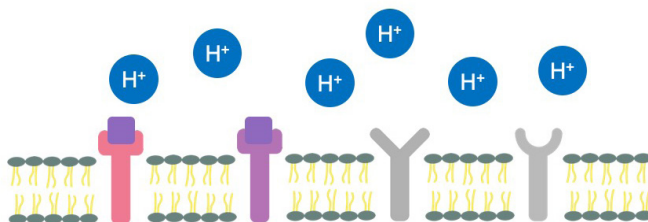
The second and main effect of vasopressin is vasoconstriction of the peripheral vasculature, which is why it can be used as a vasopressor.



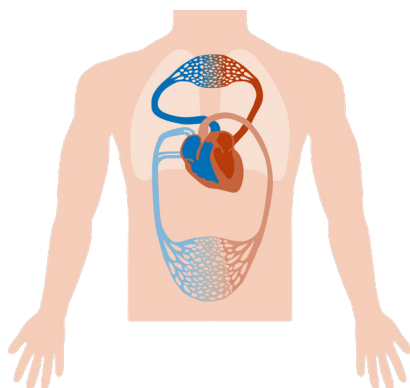
Vasopressin also has several functions as a neurotransmitter. It causes release of factor VIII and von Willebrand factor from endothelial cells, which may improve platelet function.

Vasopressin has at least two distinct advantages over other vasopressor infusions.

1. Its effect is not pH sensitive. Catecholamine receptors have diminished response in an acidic environment, which is often confounding in shock scenarios.



2. Vasopressin does not promote pulmonary vascular constriction. It may increase systemic vascular resistance without an increase in right ventricular (RV) afterload, which may exacerbate RV dysfunction.



There is some experimental evidence to suggest that vasopressin may improve brain perfusion in cardiac arrest when compared to epinephrine.

A vasopressin infusion is most commonly dosed as physiologic supplementation to augment other vasopressor therapy, in the range of 0.01–0.04 units / min.

Vasopressin may also be used at higher doses, up to 0.1 units / min, which may be useful to decrease bleeding from esophageal varices. However, at these higher doses, it carries a risk of splanchnic ischemia.

A standard dose of vasopressin bolus during cardiac arrest is 40 units IV push.

Synthetic analogs

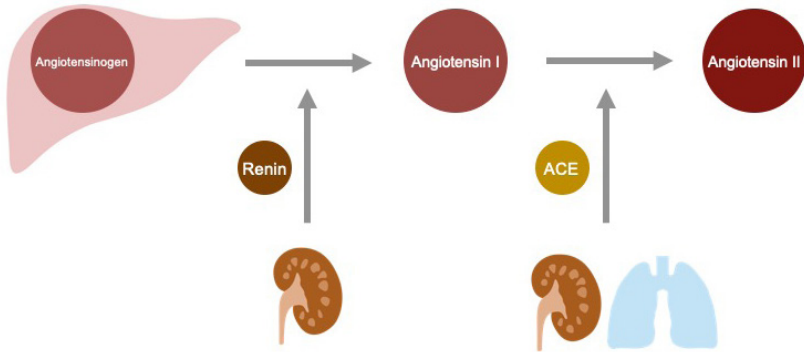
A synthetic analog, **terlipressin**, has been developed, which has increased selectivity for the V1 receptor, causing increased vasoconstriction with fewer renal effects. For this reason, it has been used in patients with hepatorenal syndrome.

Most recently, **selepressin**, has been developed with even more specificity for the V1 receptor, but clinical data supporting its use are not yet available.

Desmopressin (DDAVP) is another synthetic vasopressin analog with a longer half-life and increased selectivity for V2 receptors. Thus, it has fewer hemodynamic effects and is used more commonly for platelet dysfunction and diabetes insipidus.

Describing angiotensin II

Angiotensin II (ATII) is a naturally occurring hormone within the renin-angiotensin-aldosterone system.



Angiotensinogen is converted into angiotensin I by renin and released by the kidneys as a response to hypoperfusion. Angiotensin-converting enzyme (ACE) converts this into angiotensin II, an action which may be blocked by ACE inhibitors.

Angiotensin II increases vasopressin production from the pituitary gland as well as aldosterone production from the adrenal cortex. It also has direct venous and arterial vasoconstricting effects.

Although it was first isolated in the 1930's, it has only recently been FDA approved for therapeutic use based on data showing that it is able to elevate blood pressure in patients with vasodilatory shock.

This may be a particularly useful agent in patients with hypotension refractory to other treatments, or those who were recently taking ACE inhibitors.

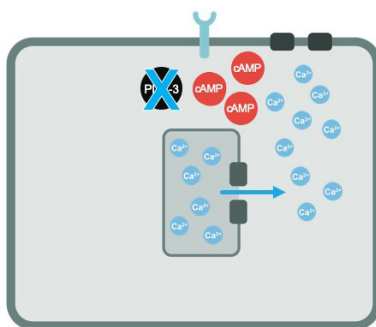
ATII increases platelet aggregation, which may increase the likelihood of thrombotic complications.

Angiotensin II is dosed as a continuous infusion, starting at $0.02 \mu\text{g} / \text{kg} / \text{min}$ ($20 \text{ ng} / \text{kg} / \text{min}$) and increased up to $0.08 \mu\text{g} / \text{kg} / \text{min}$ during the first three hours. The recommended maintenance dose should not exceed $0.04 \mu\text{g} / \text{kg} / \text{min}$.

Administering phosphodiesterase inhibitors

Milrinone is a bipyridine phosphodiesterase III inhibitor, which functions as an inodilator. Its physiologic response is similar to dobutamine but its mechanism of action is slightly different.

Milrinone inhibits the phosphodiesterase III enzyme, which normally degrades cyclic AMP (cAMP) into AMP. As a result, there are increased levels of cAMP, which stimulates calcium-induced calcium release from the sarcoplasmic reticulum, leading to myocyte contraction.



This also promotes vasodilation in vascular smooth muscle, thus hypotension is again a common effect in both the systemic and pulmonary vasculature, often more so than with dobutamine. As an inotrope and chronotrope, it is also associated with increased tachydysrhythmias.

This may be problematic in patients with low blood pressure but may be advantageous for those with high systemic vascular resistance or with high pulmonary vascular resistance.

Because of its separate mechanism of action, milrinone may be particularly useful in patients with saturated or down-regulated beta receptors, such as those already on beta-agonists, those in prolonged shock or with chronic heart failure, and those taking beta-blockers.

The typical dosing range for a milrinone infusion is 0.25–0.75 $\mu\text{g} / \text{kg} / \text{min}$.

Milrinone has a relatively long half-life of 2–4 hours, which is even longer in patients with renal failure, so the dose should be decreased in these patients.

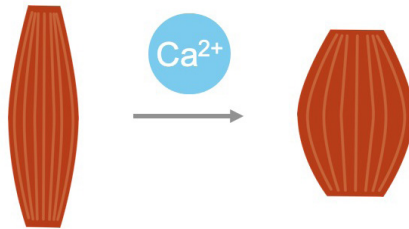
It can be used with or without a bolus (50 $\mu\text{g} / \text{kg}$), which may shorten the time of onset but is associated with more hypotension and tachycardia.

Amrinone, or inamrinone, is another phosphodiesterase III inhibitor, from which milrinone was derived. It is less potent than milrinone and causes a dose-related thrombocytopenia, so it is used less frequently.

Using calcium-based agents for shock

Calcium

Calcium plays a pivotal role in myocyte signaling and contraction. It is the final common pathway leading directly to myocyte contraction, thus manipulation of calcium concentration can have a profound inotropic effect.



Calcium by itself is most commonly available in IV formulations paired with either gluconate or chloride. The bioavailability of each appears to be similar, though calcium chloride tends to be more caustic, so a continuous infusion is preferred through a central line, when possible.

Most importantly, one gram of calcium chloride has about three times the amount of elemental calcium as a gram of calcium gluconate, so in emergent situations, calcium chloride is the preferred agent.



Calcium chloride given to treat cardiogenic shock can be dosed as a 20 mg / kg bolus, or simply a 1 g ampule followed by a 20–50 mg / kg / h infusion if there is a desirable response.

Rapid administration of calcium chloride may be associated with flushing and nausea, which is usually self-limited.

Levosimendan

Levosimendan is a calcium sensitizer, which works functionally as an inodilator.



In myocytes, it increases troponin C's sensitivity to calcium, producing increased contractility, but without increasing myocardial oxygen demand as much as catecholamine-based inotropes. It also works on ATP-dependent potassium channels, causing peripheral arterial and venous vasodilation.

While it has not been approved in the USA, it is being used successfully for heart failure exacerbations in many other countries. Side effects are similar to other inodilators including tachycardia and hypotension, plus the potential to cause hypokalemia.

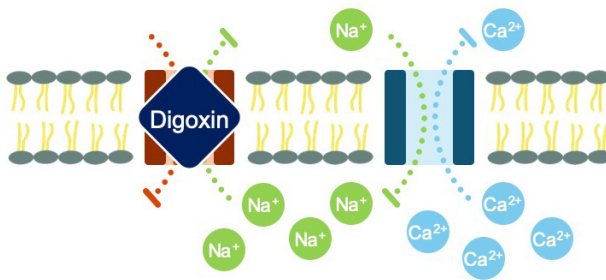
It is dosed with or without a bolus of 6–24 μg / kg, followed by an infusion at 0.05–0.4 μg / kg / min.

Levosimendan has active metabolites with a very long half-life, which may cause a prolonged effect, so infusion is recommended for less than 24 hours.

Digoxin

Digoxin is an effective rate control agent, often used for atrial fibrillation.

It inhibits the sodium-potassium ATPase, increasing intracellular sodium levels. This, in turn, decreases activity of the sodium-calcium exchange pump, thereby increasing intracellular calcium. In this way, digoxin functions as a weak inotrope.



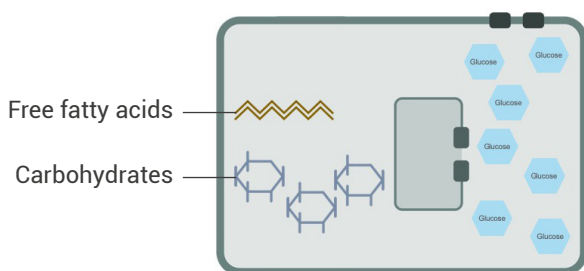
While it may not be a first-line agent for most types of shock, it may be considered if rate control is needed, as it does not have the same negative inotropic effects as other rate control agents. Its AV nodal-blocking function is not completely understood but believed to be mediated by parasympathetic stimulation.

Bolus dosing of 500 µg, repeated once or twice over 24 hours, may be considered in the short term but maintenance therapy should be adjusted to clinical effect, accounting for renal function. Signs of toxicity should be monitored.

Employing insulin and glucagon for shock

Insulin

Insulin functions as an inotrope, which was first noted shortly after its discovery in the 1920's. The exact mechanism by which this occurs is still debated. Most experts feel that by increasing the intracellular glucose concentration in the myocardium, metabolism is shifted from free fatty acids toward more efficient carbohydrate utilization.



Because this is not a catecholamine-mediated response, high-dose insulin therapy has demonstrated great efficacy for treating beta-blocker and calcium channel blocker toxicity in both animals and humans. It is unclear if this benefit translates to other patients with cardiogenic shock but it may be considered, especially for refractory cases.

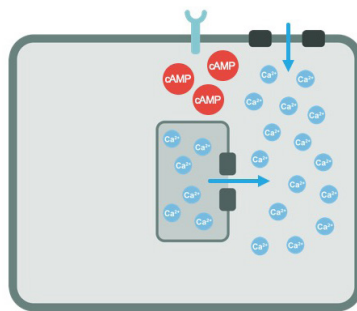
The dosing for high-dose insulin therapy includes an initial bolus of 0.5–1.0 units / kg, followed by an infusion of 0.5–1 units / kg / h.

This is likely to decrease serum glucose levels significantly, so a bolus of glucose (0.5 g / kg) is commonly included, followed by an infusion at 0.5–1.0 g / kg / h. This is often called high-dose insulin / glucose (HIG) therapy.

This may also create intracellular potassium shifts, so a solution that also includes potassium, often abbreviated GIK, has also been commonly used, especially during organ support before transplant.

Glucagon

Glucagon is a polypeptide hormone and acts as an inotropic and chronotropic agent by increasing the intracellular cAMP concentration. In this way, it functions downstream from the beta receptor and is a mainstay of treatment for beta-blocker toxicity. It may also be an effective treatment for verapamil or diltiazem toxicity.



Side effects of glucagon include nausea, vomiting, and hyperglycemia.

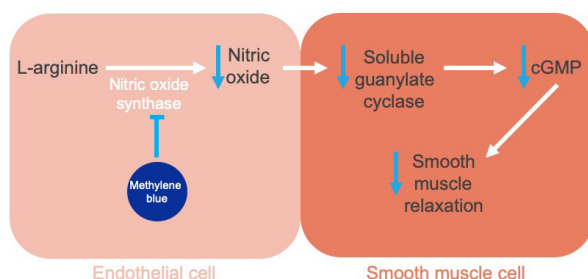
Bolus dosing is 1–10 mg and an infusion of 25–75 µg / min may be used for sustained effect.

Using methylene blue and hydroxocobalamin for shock

Methylene blue

Methylene blue is a recognized treatment for refractory distributive shock (also known as vasoplegia).

It likely works by decreasing nitric oxide production through inhibition of nitric oxide synthase and soluble guanylate cyclase.



The most notable risk with methylene blue is the development of serotonin syndrome. Although this is rare, the risk is elevated in patients taking other serotonergic medications (e.g., serotonin reuptake inhibitors or monoamine oxidase inhibitors).

Patients with a history of glucose-6-phosphate dehydrogenase (G6PD) deficiency are at risk for hemolysis from oxidative stress.

The typical dose of methylene blue is 1.5–2.5 mg / kg IV, with or without an infusion at 0.25–0.5 mg / kg / h for 4–6 hours.

Hydroxocobalamin

Hydroxocobalamin, or vitamin B12, is available as an injectable dietary supplement.

It is approved for the treatment of cyanide poisoning but has been shown to improve blood pressure in vasoplegia. It likely also works through inhibition of nitric oxide synthase.

Although it is considerably more expensive than methylene blue, it does not carry the risk for serotonin syndrome or hemolysis.

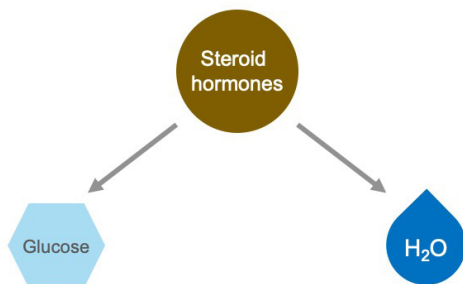
It should be noted that hydroxocobalamin may interfere with certain lab tests (e.g., hemoglobin, creatinine, and coagulation studies). It also interferes with the dialysate sensor in hemodialysis machines, usually causing an alarm.

The typical bolus dose for treating vasoplegia is 5 or 10 g over 15 minutes, which can be repeated.

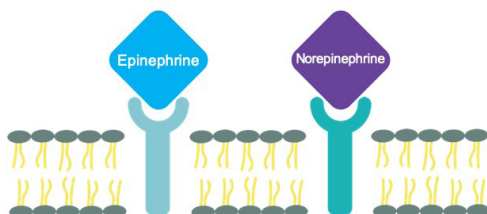
Recognizing steroids as adjuncts to vasopressor therapy

Adrenal insufficiency is an important cause of refractory shock. Even in patients without a history of adrenal problems, critical illness may unmask an inability to compensate for the added stress.

Naturally occurring steroid hormones play many important roles inducing both glucocorticoid and mineralocorticoid effects. Different agents have different durations of action, and different relative glucocorticoid and mineralocorticoid potencies.

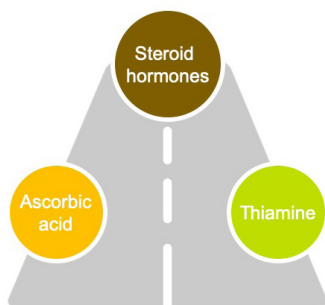


Generally, steroids influence gene expression and suppress immune responses. Glucocorticoids have a direct vasoconstricting effect and increase responsiveness to catecholamines like norepinephrine and epinephrine. They also blunt the vasodilatory effect of nitric oxide.



There is conflicting evidence around the benefits of routine steroid administration for shock but it is a reasonable consideration, especially in refractory cases or if adrenal insufficiency is suspected.

There is recent evidence that supplementation of other cofactors in the steroid pathway, specifically ascorbic acid and thiamine, may be beneficial for treating septic shock. It is reasonable to supplement these as well, especially in nutritionally depleted patients.



Hydrocortisone is the agent most commonly used to treat shock. It provides both glucocorticoid and mineralocorticoid effects.

The standard dose of hydrocortisone is 200 mg in divided doses over 24 hours or as a continuous infusion. Most evidence has been studied with a seven-day course, followed by a taper.

Relative potencies of common steroid agents

Short-acting

Name	Equivalent dose (mg)	Relative anti-inflammatory potency	Relative mineralocorticoid potency	Plasma half-life (h)	Biologic half-life (h)
Cortisone	25	0.8	2	0.5	8–12
Hydrocortisone	20	1	2	1.52	8–12

Intermediate-acting

Name	Equivalent dose (mg)	Relative anti-inflammatory potency	Relative mineralocorticoid potency	Plasma half-life (h)	Biologic half-life (h)
Methylprednisolone	4	5	0	1.5–3	18–36
Prednisone	5	4	1	1	18–36
Prednisolone	5	4	1	2–3.5	18–36
Triamcinolone	4	5	0	3.5–4	18–36

Long-acting

Name	Equivalent dose (mg)	Relative anti-inflammatory potency	Relative mineralocorticoid potency	Plasma half-life (h)	Biologic half-life (h)
Betamethasone	0.6	20–30	0	5.5	36–54
Dexamethasone	0.75	20–30	0	2–3.5	36–54

Choosing parasympatholytic agents wisely

A decrease in cardiac output or blood pressure may result from excessive parasympathetic stimulation from various stimuli (e.g., pain or emotional distress) or seemingly insignificant events like turning in bed or suctioning an endotracheal tube. While these are most commonly self-limited, they may precipitate a cardiac arrest or may be recurrent.

If excess parasympathetic activity contributes to a shock state, a parasympatholytic agent may be useful.

Atropine

Atropine is a belladonna alkaloid derived from the nightshade family of plants. It was first identified in 1833. It is a muscarinic antagonist with strong anticholinergic properties.



Atropine may be given as a 0.5–1 mg IV bolus, which will usually last about 30–60 minutes and may be repeated.

Side effects include tachydysrhythmias, vision changes, dry mouth, and mental status changes, especially in elderly patients. At very low doses atropine may cause a paradoxical decrease in heart rate (presumably by a central vagal response) and it is not recommended for use in patients who have received a heart transplant, due to the risk of prolonged heart block.

Glycopyrrolate

Glycopyrrolate, or glycopyrronium, is a synthetic derivative of atropine, with the advantages that it is longer lasting and does not cross the blood brain barrier, thus it does not have an effect on mental status.



It may be dosed in 0.1 mg boluses every few minutes (titrated to effect), although commonly a dose of 0.2 or 0.4 mg may be used and should last 2–3 hours.

Chapter 3

SPECIAL CLINICAL SITUATIONS



Managing cardiogenic shock

Cardiogenic shock is inadequate perfusion originating from failure of the heart to pump blood effectively.

The cardiac output produced by the heart is equal to the stroke volume times the heart rate.



A low heart rate may be a cause of decreased cardiac output, or may be a contributing factor if a compensatory tachycardia is blunted. Chronotropic agents may be useful if there is a reversible cause. These may include isoproterenol, dobutamine, milrinone, epinephrine, glucagon, calcium or dopamine.

The preferred agent depends on whether there is alternate pathology present (e.g., ischemia, hyperkalemia or a nodal-blocking toxin). Isoproterenol is likely the most effective agent from the standpoint of purely increasing the heart rate, but it is associated with increased cost and risk of hypotension.

While increasing heart rate may improve cardiac output, increasing stroke volume may also have a profound effect.

A low stroke volume may be the result of a variety of different conditions including ischemia, cardiomyopathy, hypocalcemia or the presence of a toxin. There are also mechanical considerations (e.g., hypovolemia or obstruction), which may decrease the stroke volume.

The options for inotropic medications are essentially the same as chronotropic agents, each with relative advantages and disadvantages.

Dobutamine is likely the first-line agent for inotropy for most providers.

For patients with high blood pressure, pulmonary hypertension or those taking beta-blockers or beta-agonists, milrinone may be the most effective agent.

For patients with a low blood pressure, an agent with combined inopressor effects (e.g., epinephrine) may be preferred.

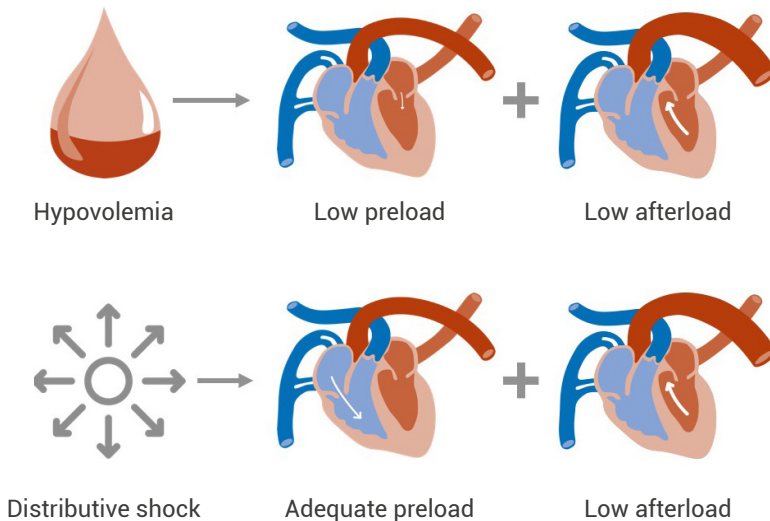
Low cardiac output as a result of hypocalcemia, adrenal insufficiency, hypovolemia, obstruction or toxic ingestion should obviously have the underlying cause addressed.

Mechanical circulatory support may be considered as a temporary measure to support the patient with cardiogenic shock if a reversible cause (e.g., coronary occlusion or toxic overdose) is present.

Treating distributive shock

Distributive shock occurs when there is inadequate vascular resistance resulting from vasodilation. This is very closely related to hypovolemia, since there is inadequate volume relative to the size of the vasculature.

An assessment of preload may help with this distinction, either by point-of-care echocardiography, pressure monitoring or some other means. Both distributive or hypovolemic shock are likely to have a low afterload, which may cause a hyperdynamic left ventricle. However, hypovolemia is likely to also have a low preload, while purely distributive shock may have adequate preload.



IV fluids may be an initial step in management and may address concomitant hypovolemia; however, vasopressors directed at increasing vascular resistance will address the physiology directly.

It is important to remember that enhancing perfusion (not pressure) is ultimately the goal. Raising pressure with vasoconstrictors, without increasing flow, may not improve perfusion. However, they may allow blood flow to be redistributed toward more vital organs if used appropriately. For example, coronary perfusion relies heavily on diastolic pressure and may improve with peripheral vasoconstriction from a vasopressor.

Phenylephrine—causes pure systemic and pulmonary vasoconstriction, which has the potential to decrease cardiac output by increasing afterload.

Norepinephrine—adds some inotropy via mild beta 1 agonism and is generally the preferred agent in undifferentiated shock.

Vasopressin—increases systemic but not pulmonary vascular resistance. Unlike the other agents, it is equally efficacious at a low pH, which is often present with shock.

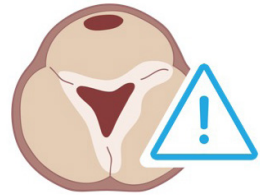
Angiotensin II—is a newer agent targeted at increasing vascular resistance through the renin-angiotensin system. It may be useful as an adjunct for distributive shock, especially in patients resistant to other agents or those taking ACE inhibitors.

Methylene blue or hydroxocobalamin—this may be considered in especially refractory cases of distributive shock, to increase perfusion pressure.

Remember that some agents (e.g., dobutamine and milrinone) may also decrease systemic vascular resistance, so sometimes decreasing the dose or adding an additional vasopressor may be necessary.

Handling valvular disease

Abnormalities in the heart valves may have profound hemodynamic effects. These may be classified as cardiogenic shock but management differs from pure systolic dysfunction.



Definitive treatment for these lesions is almost always surgical but medical therapies may help improve performance and support perfusion, while awaiting definitive repair.

In general, stenotic lesions (e.g., mitral stenosis or aortic stenosis) tend to limit filling times, so tachycardia is not well tolerated when these types of lesions are present. Atrial fibrillation tends to create a particularly unstable response, so it is prudent to avoid chronotropic agents, which may increase heart rate or promote tachydysrhythmias.



Mitral stenosis

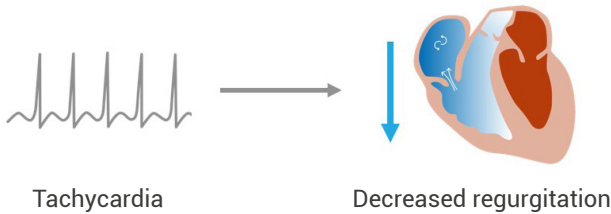


Aortic stenosis

For aortic stenosis in particular, inotropic agents may exacerbate the problem by adding increased myocardial demand. Any additional force of contraction is limited by the fixed obstruction, so there is generally no increase in cardiac output.

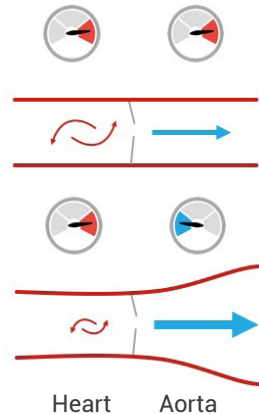
Rate control agents or cardioversion may be necessary for tachyarrhythmias with stenotic valvular disease. If diastolic perfusion pressure is inadequate, vasopressors like phenylephrine or vasopressin, which do not promote inotropy or chronotropy, would be preferred.

On the other hand, regurgitant lesions may provide better flow at higher heart rates, as there is less time for regurgitant flow.



Positive inotropic and chronotropic agents like milrinone or dobutamine, may facilitate forward flow and perfusion.

Some hypotension may be tolerated in these scenarios as well, as this will facilitate a *down-stream* pressure gradient, promoting forward flow.



For stenosis—target lower heart rate and higher afterload.

For regurgitation—target higher heart rate and lower afterload.

Managing right ventricular dysfunction

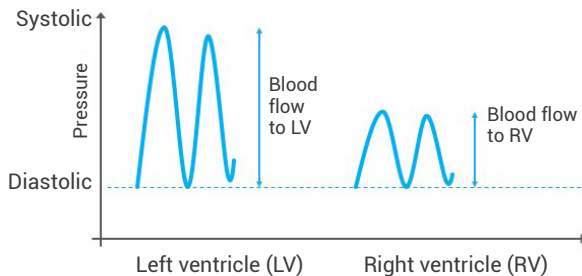
Right ventricular (RV) failure is just as important as left ventricular causes of heart failure, though it is often less recognized. Stroke volume and cardiac output through the right heart must match that of the left heart, except in cases of extreme shunting.



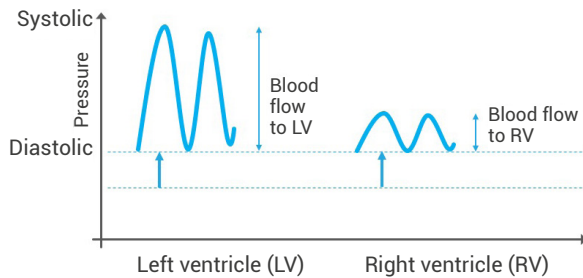
However, there are several ways in which the right heart has a more fickle physiologic equilibrium.

The right side of the heart is a much lower pressure system, thus the right ventricle is much more sensitive to increased afterload. This may result from chronic conditions like left-sided heart failure or lung disease and may be exacerbated acutely by volume overload, hypoxemia, hypercarbia, acidosis or positive pressure ventilation.

The left ventricle is perfused during diastole but the right ventricle has a smaller pressure gradient. It relies on perfusion throughout the cardiac cycle.

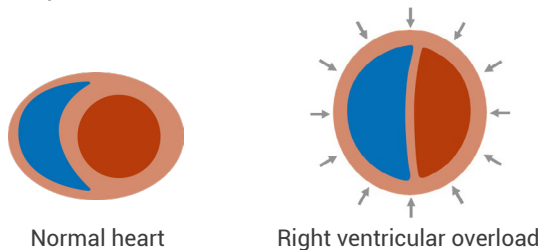


Therefore, a small increase in the right ventricular wall tension can have dramatic effects on right ventricle perfusion.



Increased right ventricular pressure also increases regurgitation through the tricuspid valve, further limiting forward flow.

Increased pressure in the right ventricle will also distort the normal architecture of the heart, decreasing the efficiency of right ventricular contraction and shifting the intraventricular septum to the left. This, in turn, limits left ventricular filling and stroke volume and is known as ventricular interdependence.



If possible, reversing underlying causes of pulmonary hypertension should be addressed. This may include the use of pulmonary vasodilators or pulmonary artery catheters in some patients, which should be done with expert consultation. Transfer to a specialized center or one capable of mechanical circulatory support should be considered early for patients with right-heart dysfunction.

Positive pressure ventilation—should be avoided, if possible, although oxygenation and ventilation should be optimized, so this may be a tenuous balance.

Diuresis—may be helpful to decrease filling pressures, however, hypovolemia may also decrease cardiac output. If fluid boluses are necessary, they should be small with frequent reassessment.

Inotropic agents may be particularly useful with right ventricular dysfunction but attention should be paid to their additional effects.

Dobutamine and milrinone—may increase inotropy but usually decreases blood pressure, which may compromise RV perfusion.

Vasopressin—vasopressors may help maintain perfusion pressure (especially vasopressin), which will not increase pulmonary vascular resistance and may in fact lower it.

Norepinephrine and epinephrine—remain viable options, depending on the degree of inotropy and vasopressor activity needed.

Phenylephrine—should generally be avoided for right ventricular dysfunction because it increases right ventricular afterload and does not support inotropy.

Non-sinus rhythms are not well tolerated in the setting of right ventricular dysfunction because the loss of atrial contraction decreases the cardiac output. Atrial fibrillation should generally be cardioverted immediately, particularly if the patient is unstable.

Treating diastolic dysfunction

Diastolic relaxation is an active process, which is as important to cardiac output as systolic contraction. Patients with poor systolic function most often have diastolic dysfunction but diastolic dysfunction may also exist with a preserved ejection fraction.



Diastolic function $\neq$ systolic function

Common causes of impaired diastolic relaxation

- Fibrosis
- Hypertrophy
- Genetic mutations
- Electrolyte changes

The severity of diastolic dysfunction is most commonly assessed by echocardiography and graded from 0–4.

Grade 0 Normal	Grade 1 Impaired relaxation	Grade 2 Pseudonormal	Grade 3 Restrictive (reversible)	Grade 4 Restrictive (irreversible)
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Small changes in preload and afterload have more dramatic effects in hearts with diastolic dysfunction. Improvements may be noted by decreasing ventricular filling pressure if it is high (e.g., diuresis

or lowering blood pressure) or increasing it if it is low (e.g., volume replacement or vasopressor therapy).

Ischemia has a detrimental effect on diastolic function, which may decrease cardiac output. Attempts should be made to optimize perfusion and decrease myocardial oxygen demand (e.g., through sedation or pain control).

Positive pressure ventilation may decrease preload, decrease afterload, and provide improved oxygenation and decreased work of breathing, all of which are likely to improve diastolic function, especially in a patient with excessive intravascular volume.

Tachycardia impairs diastolic filling time. Agents to slow the heart rate may be useful. However, these agents are also negatively inotropic and may impair systolic function. Conversely, positive inotropes tend to increase heart rate and limit diastolic filling time.

Beta-adrenergic agonists like epinephrine and dobutamine may help promote diastolic relaxation as long as filling time is adequate.

Calcium reuptake into the sarcoplasmic reticulum plays an important role in diastolic function and electrolytes should be kept normal. Levosimendan may help promote inotropy through calcium release and appears to improve diastolic function. However, this benefit may be limited by the presence of tachycardia.

Augmenting mechanical circulatory support

Mechanical circulatory support may help augment ventricular function. This may be available for the right ventricle, the left ventricle or both. These may be temporary or more permanent.



Intra-aortic balloon pumps (IABPs)—have been used for many years to provide support during cardiogenic shock. They augment diastolic filling pressure by expanding in the aorta during diastole. In general, they provide about 1–2 L / min of augmented cardiac output.

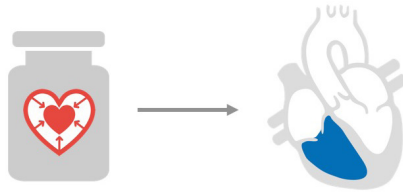
Percutaneous ventricular assist devices—such as Impella® and Tandem Heart® have been developed more recently to provide more flow (up to several L / min) to support cardiac output.

Extracorporeal membranous oxygenation (ECMO) devices—are able to provide biventricular support (with a venous-arterial configuration) at higher flow rates, and can also oxygenate blood to augment the function of the lungs.

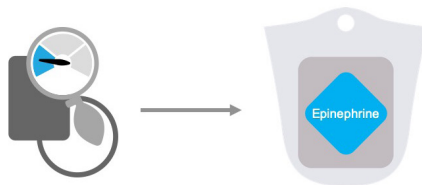
Cardiopulmonary bypass (CPB)—is able to replace the heart and lung functions completely but is generally available only in the operating room.

Although mechanical devices can augment flow, patients often need additional support for blood pressure with vasopressor agents (e.g., norepinephrine or vasopressin).

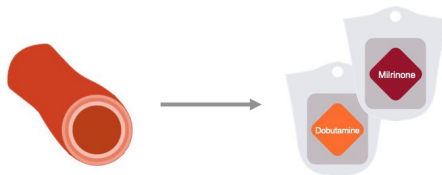
Patients may also require inotropic support, especially if only one side of the heart is being supported mechanically. Most commonly this would mean an inotrope like dobutamine to support right ventricular function, while the left ventricle is supported with a device like an IABP or Impella®.



If the blood pressure is low, an inopressor like epinephrine may be advantageous.



If vasodilation is desired, an inodilator like milrinone or dobutamine may be preferred, particularly for right-heart support in patients with a higher blood pressure.

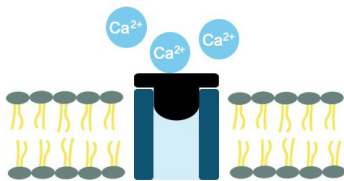


It is important to remember that mechanical circulatory support provides temporary assistance as a bridge for patients in whom an endpoint is clear. For example, a disease process in which recovery is expected or definitive treatment is available (e.g., organ transplant).

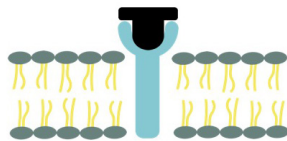
Coping with toxic overdose

The pathway for myocyte contraction can be interrupted in several places.

The most common significant cardiotoxic overdoses are non-dihydropyridine calcium channel blockers (e.g., diltiazem, verapamil) and beta-blockers (e.g., metoprolol, atenolol).



Calcium channel blocker



Beta-blocker

Calcium channel blocker toxicity

Calcium channel blockers interrupt the influx of calcium at the cell membrane, which can decrease heart rate and the strength of contraction.

This may be overcome by competitive agonism with aggressive calcium chloride administration.

In addition, glucagon may help create an alternative pathway for myocyte contractility.

Epinephrine or other inotropes or vasopressors may be considered as adjunctive therapies as well.

For refractory cases, high-dose insulin and glucose infusion or intravenous lipid emulsion may be considered. Ultimately, mechanical circulatory support may be considered for recoverable cases.

Beta-blocker toxicity

Beta-blockers on the myocyte also block the influx of calcium through the calcium channel and blunt the increase in intracellular cyclic AMP.

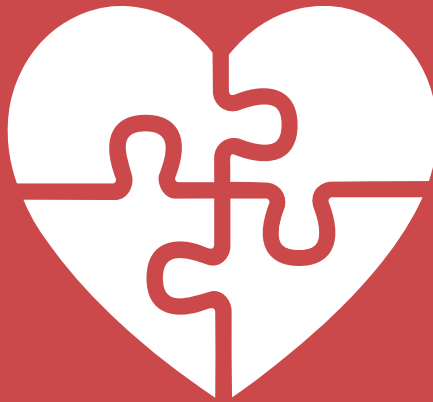
Again, this may be bypassed with the use of glucagon. Calcium supplementation may also be helpful in overdose.

Beta-agonists like isoproterenol, dobutamine or epinephrine may help offset the receptor blocking effect.

High-dose insulin and glucose, lipid emulsion or mechanical circulatory support are again considered for refractory cases.

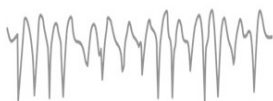
Chapter 4

PUTTING IT TOGETHER



Approaching undifferentiated hypotension

An organized approach to a patient with shock can help direct interventions toward improving perfusion. By evaluating cardiac pump function, preload, and afterload, the source(s) of derangement can quickly be identified and addressed, even if the cause of shock is not yet identified.



Ventricular dysrhythmia



Valvular abnormality

Causes of cardiogenic shock may be obvious (e.g., ventricular dysrhythmia) or may be more cryptic (e.g., valvular abnormality). A point-of-care ultrasound examination can quickly identify severe valvular abnormalities and can identify if the contraction is hyperdynamic, normal or depressed.

If there is a significantly depressed systolic function (low ejection fraction), an agent with inotropy (epinephrine or dobutamine) is much more likely to be effective in restoring adequate cardiac output than a vasoconstrictor alone.

If the heart is normal or hyperdynamic, then the question becomes whether there is inadequate preload, afterload or both. Again, ultrasound is one way to establish that there is adequate preload to the ventricles.

Again, it is useful to identify whether the problem is related to the left or right side of the heart or both.

Once adequate preload and cardiac function are established then afterload is the only problem that remains and a vasoconstrictor (e.g., norepinephrine or phenylephrine) can help augment vascular resistance.

In the same fashion, we can identify which of the four types of shock are present.

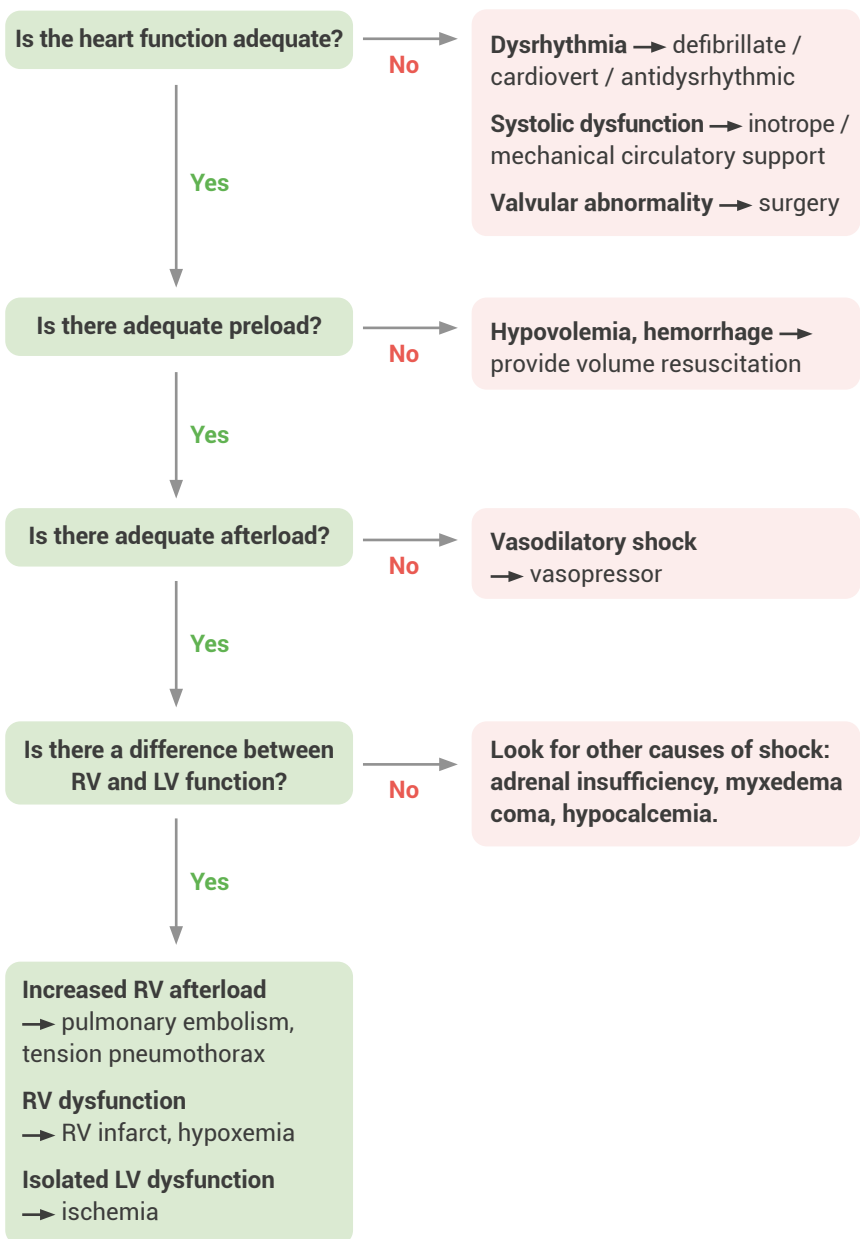
Cardiogenic shock—results in inadequate pump function, depressed systolic function or valvular abnormalities.

Hypovolemic shock—results in inadequate preload and underfilled ventricles or vasculature.

Distributive shock—results in inadequate afterload and hypotension despite normal heart and filling pressures.

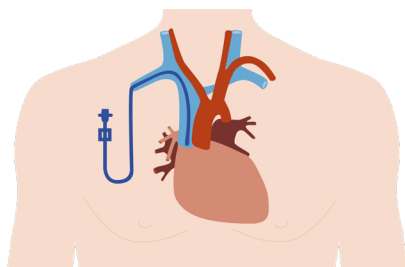
Obstructive shock—may cause any of these, depending on where the lesion is. Compression of the vena cava may cause inadequate right ventricular preload, while compression of the heart may cause inadequate pump function.

If the lesion is between the right and left ventricles (e.g., pulmonary embolism), then preload to the right ventricle will seem adequate with or without good contractility, while the left ventricle will usually appear underfilled and may be hyperdynamic.



Administering vasopressors

The most common method for administering inotropes and vasopressors is a continuous infusion through a central line. This ensures intravascular administration and decreases the likelihood for extravasation. However, it brings with it the risk of complications associated with central line placement and may cause delay, as it takes time to place.



Central line

Administration of vasoactive agents through a peripheral IV is becoming increasingly established. This may be considered, especially in the short term and with lower dose infusions or while central access is being established.



Peripheral infusion

Agents like phenylephrine and epinephrine are approved for subcutaneous administration, so even if extravasation occurs it is rarely a problem. However, agents like norepinephrine and dopamine may cause complications if there is extravasation. Fortunately, significant complications from vasoactive extravasations remain rare and the benefits of peripheral administration must be considered in each scenario as well.

Traditional dosing during cardiac arrest is 1 mg of epinephrine or 40 units of vasopressin as an IV bolus, but these extremely large doses may not be ideal for many patients.

Another way to administer vasoactive agents is through small boluses, often called *push doses*. Most commonly, these would consist of a diluted concentration of a vasoactive agent (e.g., 10 µg / mL epinephrine or 100 µg / mL phenylephrine), administered in small doses (e.g., 1–2 mL at a time).

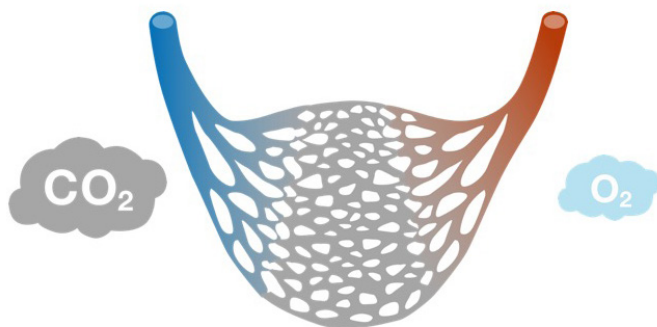
Assessing the effectiveness of your interventions

It is important to assess the efficacy after an intervention in order to determine if it was successful. Assessing tissue perfusion can be challenging.

Serum lactate can give a global estimate of tissue perfusion but it may be confounded by regional variation, other sources of lactate production like epinephrine or by multiple concomitant processes (as occurs in sepsis).

It may be useful to look at surrogates for organ perfusion like the rate of oxygen extraction, as measured by a central venous or mixed venous blood gas.

Arteriovenous differences in carbon dioxide can give estimates of oxygen utilization at the tissue level since more carbon dioxide is present on the venous side when there is inadequate perfusion. Methods to measure this are not ubiquitous.



Cardiac output can be a useful estimate of blood flow in the larger vessels and can be measured through several means. However, it does not guarantee adequate oxygen delivery at the microvascular level and must be interpreted in the context of how much blood flow is needed for a specific scenario.

Non-invasive cardiac output monitors (NICOM) and peripheral perfusion index devices may also give an idea about flow but also do not determine if this flow is adequate for perfusion.

Looking for a positive change after an intervention is the only way to assess if the intervention was successful. This may be an increase in blood pressure or cardiac output or improvement in organ function.

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